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Aurigene Pharmaceutical Services Limited

Case Study: Development of a Novel Biotherapeutic Protein for regeneration of Target Tissue

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Key Highlights

- First in class, novel large molecule
- *E. coli* expression (expressed in inclusion body)
- Successful proof-of-concept in in vitro and in vivo models
- In vivo PK, tox (non-GLP) in rodents
- Process, analytical and bioanalytical method development
- Generation of GMP material Phase I clinical DP (Ongoing)
- GLP tox to be initiated

Resource utilized

Exploratory work: Lab scale non-GMP production, in vitro and in vivo studies, non-GLP tox and gram scale material generation – 9 months, average 5 FTEs

Development work: Up to Phase I material generation, currently ongoing in FFS mode 915 months projected time

SDS-PAGE and Mass spectrometry analysis

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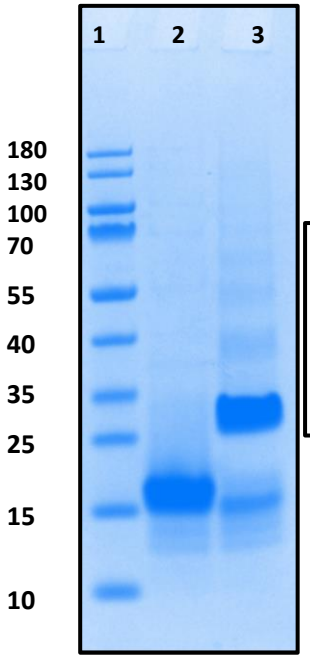
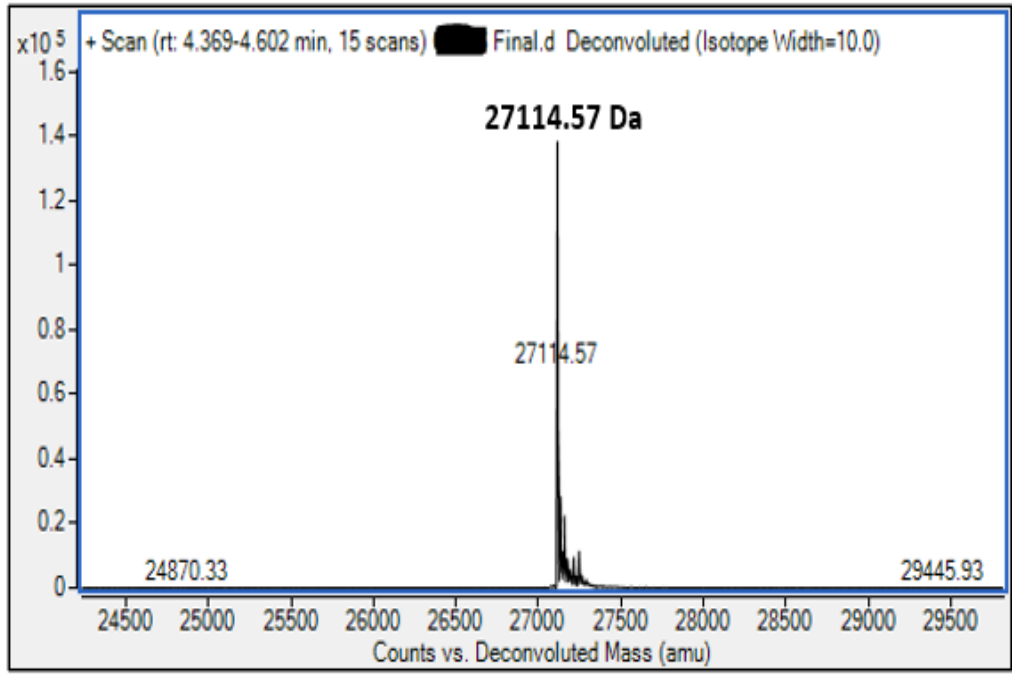


Lane 1: Marker

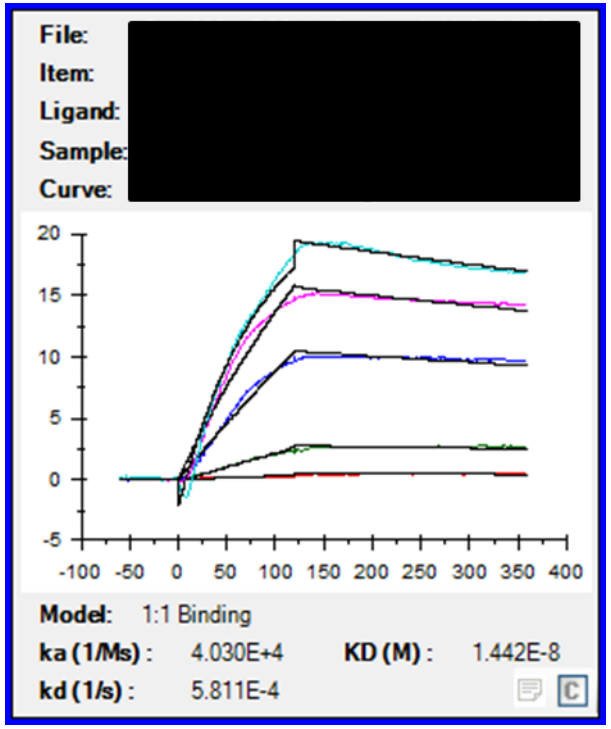
Lane 2: Reduced

Lane 3: Non reduced

SDS-PAGE



Mass Spectrometry



SPR study

Bioanalysis (SPR based)

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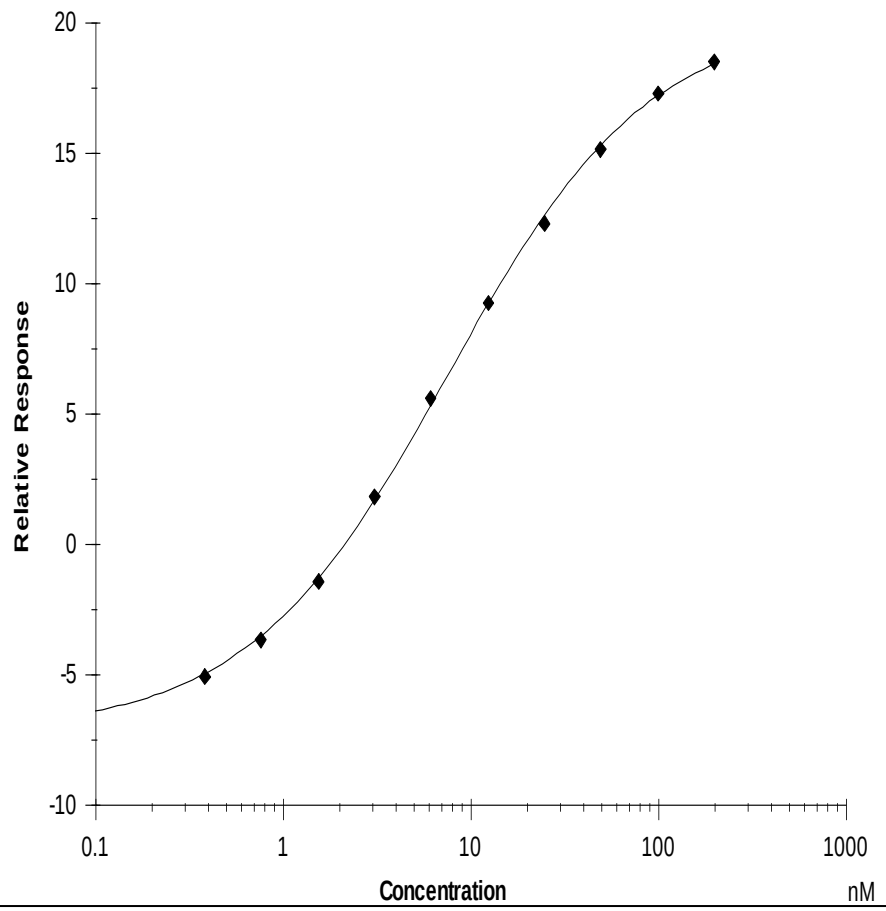
Standard

Conc. (nM)	Calc. Conc. (nM)	% Recovery
0	0.0151	NA
0.39	0.3566	91
0.78	0.7347	94
1.56	1.517	97
3.12	3.225	103
6.25	6.523	104
12.5	12.68	101
25	23.43	94
50	47.64	95
100	104.2	104
200	219.1	110

QC

Spike Conc. (nM)	Calc. Conc. (nM)	% Recovery
1	1.046	105
5	4.481	90
50	48.14	96
100	101.1	101

Standard Curve



% Recovery for BCC of Standards and QC's were between 80 – 120%

In vitro functional test

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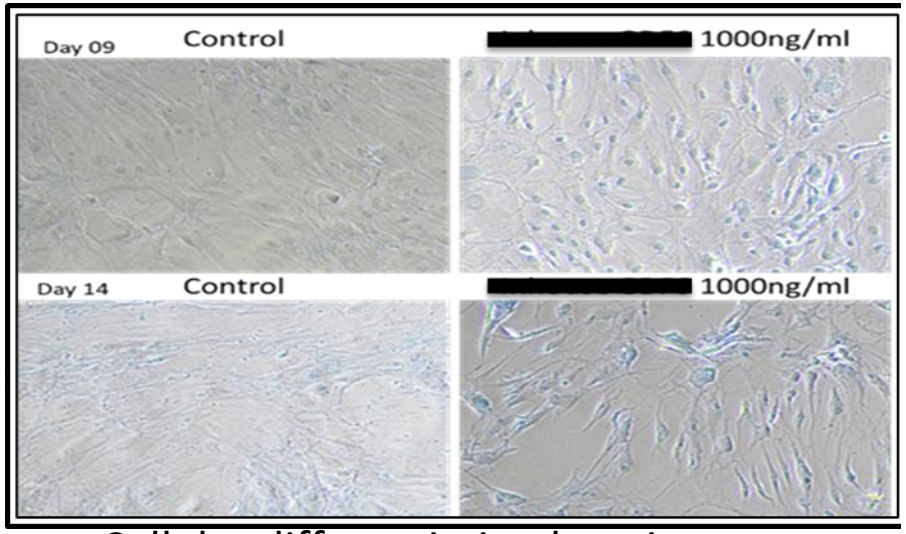
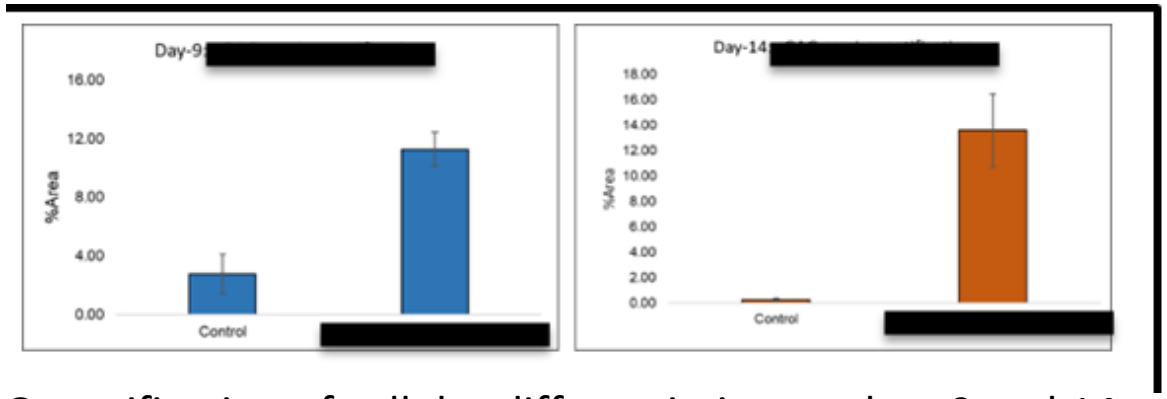
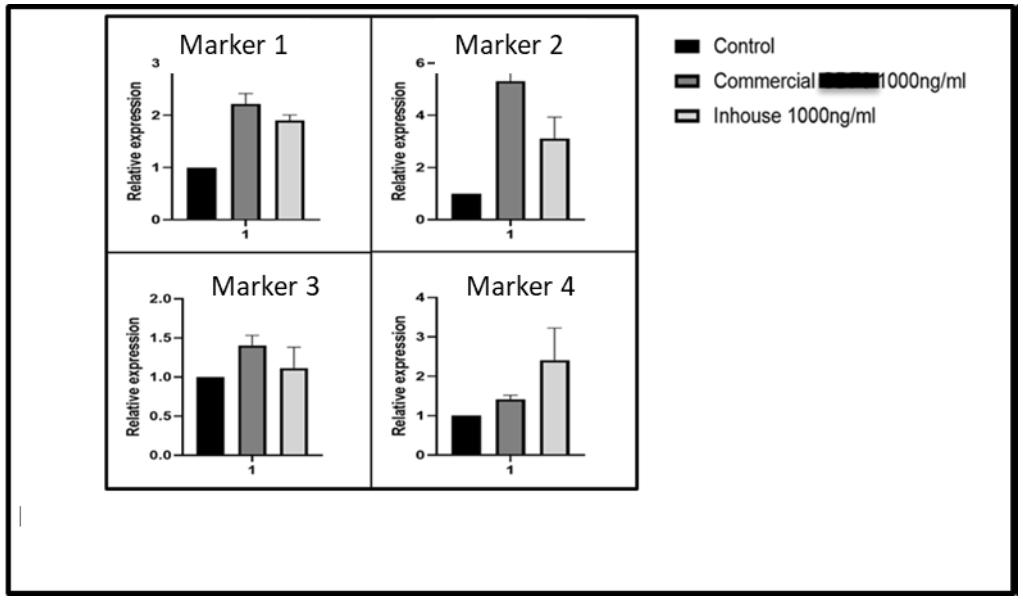


Cellular differentiation by microscopy



Quantification of cellular differentiation on days 9 and 14

Expression analysis of biomarkers (qPCR)



IHC sowing in vivo differentiation of muscle tissue (normal rodent)

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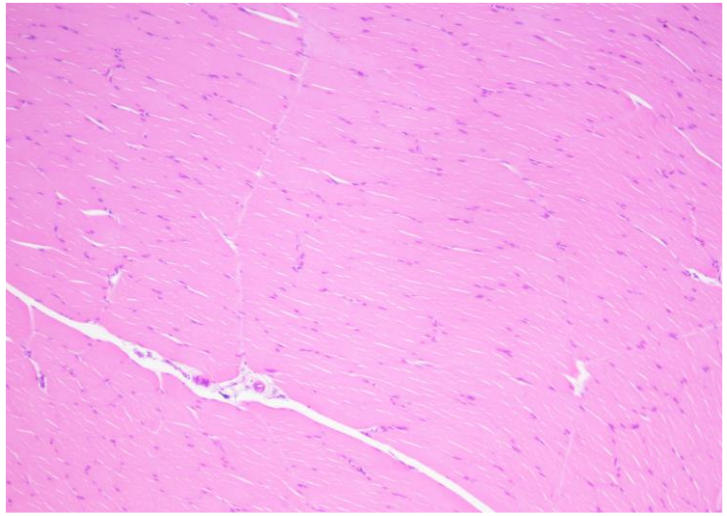
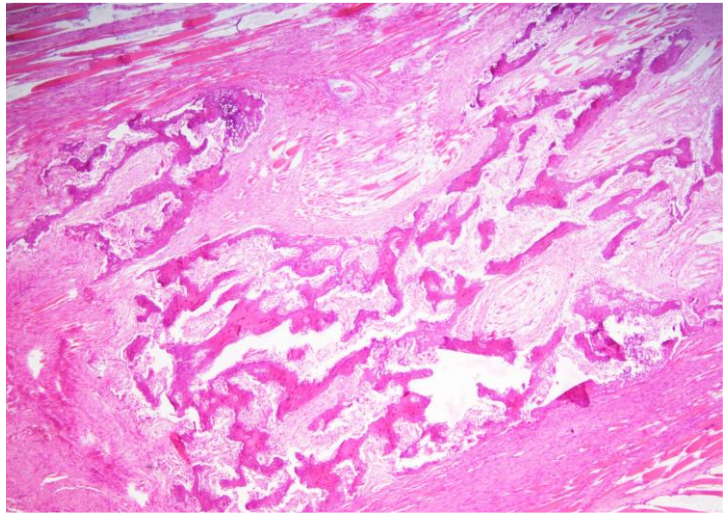
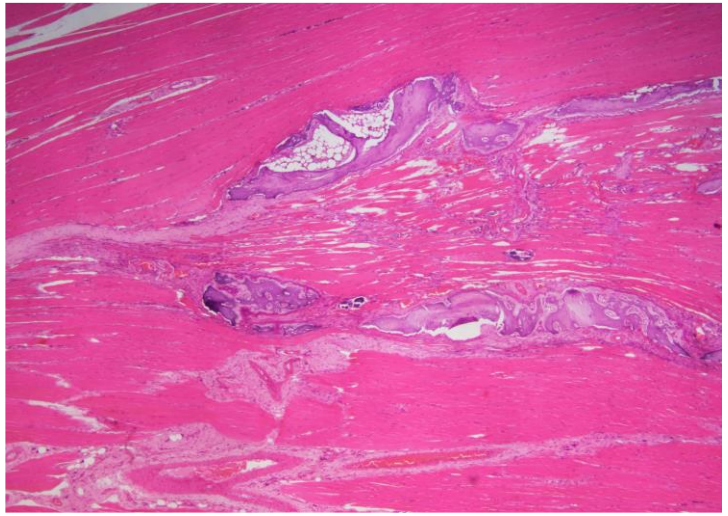


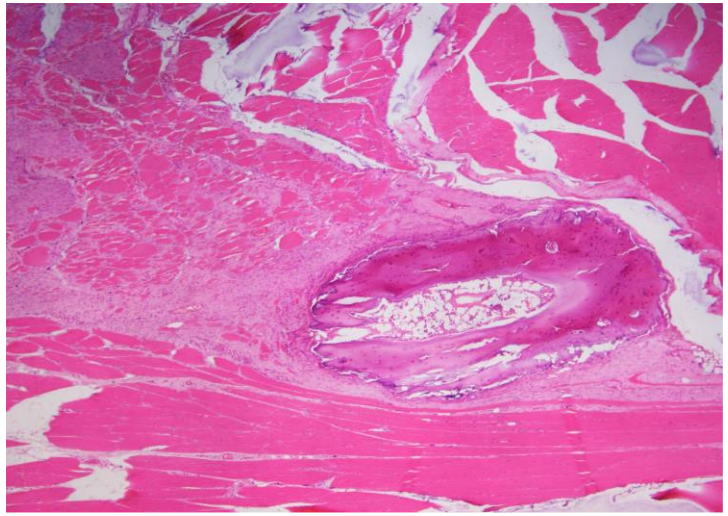
Normal



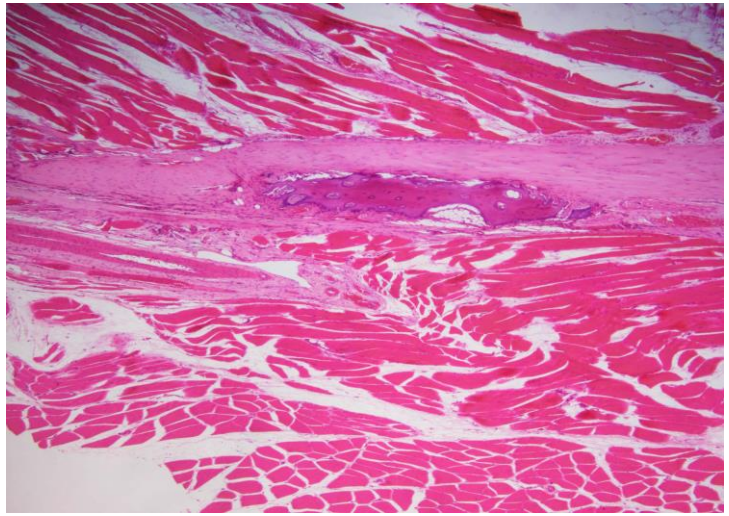
Treated 2 weeks



Treated 4 weeks



Treated 6 weeks



Treated 8 weeks

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